Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI / ML engineer shipping production healthcare AI end to end: HIPAA-conscious RAG (~35% retrieval precision, >90% grounded alignment), predictive ML pipelines (15-20% recall on high-risk patient categories at >90% precision), agentic LLM workflows (>30% hallucination reduction), and FastAPI + Docker packaging (~30% post-deploy defect reduction). Production computer vision via YOLOv8 (~30% inference latency reduction). Master's in Computer Science, Kent State. Targeting Daxko's AI Software Engineer I seat - production RAG pipelines (recursive semantic chunking + transformer embeddings + grounding controls in healthcare workflows directly map to Daxko's RAG ingestion + chunking + embeddings + vector search charter), Manga Lens 4-provider AI vision integration shipped solo on Chrome Web Store with cost-aware payload routing (proves third-party model API integration at production quality, 1:1 with Daxko's OpenAI / Azure OpenAI / Bedrock / SageMaker integration scope), Agentic Healthcare Claims with schema-validated JSON contracts between five agents (matches Daxko's reusable inference + workflow automation infrastructure), and FastAPI + Docker microservices with structured logging (matches REST API + microservices + CI/CD requirements). US Remote, F-1 OPT US-base.

CORE COMPETENCIES

Applied AI Engineer • RAG / Embeddings / Vector Search • Multi-Provider LLM API Integration (OpenAI, Anthropic, Azure OpenAI) • Agentic LLM Workflows / Schema Contracts • Python / TypeScript • LangChain / LangGraph / LlamaIndex • FastAPI / Flask / Docker / REST • CI/CD (Jenkins, GitHub Actions) • HIPAA-conscious Data Governance • US Remote (F-1 OPT)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc. *Jul 2025 - Present*

Jr. AI/ML Engineer Trainee - Applied AI, Healthcare

USA - Healthcare Technology, Applied AI

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved retrieval precision **~35%**; retrieval-grounded response alignment **>90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; agent response stability **~25%**; hallucinations **>30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement scoring, support prioritization; recall **+15-20%** on high-risk categories at **>90%** precision via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects **~30%**.
- EHR + appointment + ticket preprocessing (Pandas, NumPy); dataset reliability **>98%**, downstream instability **-40%**.
- Authored **HIPAA-conscious data governance** and stakeholder-facing system-limitation docs; ran weekly stakeholder calls with clinical SMEs.

Energy Solutions International (Emerson) *Jun 2022 - Apr 2023*

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP

- Optimized SQL and **T-SQL stored procedures** on the Synthesis Order-to-Cash platform; query execution time **~20%**, data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; deployment errors **>30%**, release cycle **~35-40%**.
- Designed **SQL Server DMVs and Grafana** performance dashboards for long-running queries, deadlocks, CPU/IO contention; incident recurrence **~25%**, RCA speed **~30%**.
- Implemented **RBAC and audit logging for financial modules** in a compliance-sensitive oil & gas environment.

Suvidha Foundation *Jan 2022 - Mar 2022*

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization + timestamp-aligned clip extraction cut manual review **60-70%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask** API with a lightweight web interface for non-technical staff.

PROJECTS

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped, Chrome Web Store)

Real-time manga / webtoon translation extension shipped to the Chrome Web Store. **Multi-provider abstraction layer** integrating four AI vision providers (Claude Sonnet, GPT-4o-mini, GPT-4.1-Nano, Ollama/minicpm-v) with per-provider payload handling (WebP for cloud, JPEG for local), cost-aware routing, circuit-breaker fallback. Manifest V3 service workers; multi-section capture pipeline for tall panels with coordinate remapping; 7-day cache; per-domain selector configs for 29 manga/webtoon sites; narrowed host permissions and privacy policy.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama, multi-provider integration

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Five-stage multi-agent pipeline (Intake Normalization, Validation, Consistency Analysis, Duplicate Detection, Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring with reasoning traces.

Python, LangChain, FastAPI, vector search, GPT-4, schema contracts

Dream Decoder - AI Multimodal Creative Intelligence

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar interpretation + Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** - **+30%** contextual alignment and **+25-30%** first-pass image success rate over naïve direct-prompt orchestration.

FastAPI, React, TypeScript, structured prompt schemas, multimodal orchestration

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced two-stage CNN pipeline with a unified **YOLOv8** detection-and-classification model - inference latency **~30%**. Sliding-window confidence aggregation suppressed blink-driven false positives **~25%**; adaptive frame skipping + NMS tuning for stable real-time operation via OpenCV video capture.

PyTorch, YOLOv8, OpenCV, real-time inference

EDUCATION

Master of Science, Computer Science - *Kent State University* *Aug 2023 - May 2025*

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - *KL University* *Jun 2019 - May 2023*

Founder / Lead of E-Farming platform (university project).

CERTIFICATIONS

Deep Learning Specialization - *DeepLearning.AI (Coursera)* *2024*

SKILLS

LLM / GenAI: RAG (chunking, embeddings, vector search), agentic workflows, structured outputs, evaluation pipelines, guardrails, grounding, LangChain, LlamaIndex, multi-provider AI integration (OpenAI, Anthropic, Ollama)

Languages: Python (primary), TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), C++

ML Frameworks: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, threshold calibration

Backend / Inference: FastAPI, Flask, REST, Docker, structured logging, load simulation, microservices

Frontend: React, TypeScript, Vite, Tailwind

DevOps: Docker, Jenkins, GitHub Actions, Grafana, CI/CD, cloud-ready deployment

Data Governance: HIPAA-conscious de-identification, data lineage, evaluation audit trails, RBAC

Domains: Applied AI, Healthcare AI, Workflow Automation, Multi-Provider AI Integration